Project Description

1. **Introduction:** This paper outlines a set of research proposals that aim to investigate the formal properties and processing mechanisms underlying sentences in which a single Wh-filler is associated with multiple gap sites. In addition to sentences containing subject (1a) or adjunct (1b) parasitic gaps (PGs), this project also examines prepositional dative (1c) and complement clause (1d) PGs, as well as so-called *symbiotic gap* structures (1e; Levine & Sag 2003).

(1) a. Who did a friend of __ recommend __ for the job?
b. What did Mary write a review for __ without reading __?
c. Who did Mary introduce an admirer of __ to __ yesterday?
d. Who did John reassure __ that he would speak with __ next week?
e. What kinds of books do authors of __ hate creating the index after writing __?

There are two main goals that will be pursued in this research program. I intend to show that (i) during online processing the parser predictively builds syntactic representations containing PGs; and (ii), the licensing of PGs can be accounted for within a Minimalist Program (Chomsky 1995) framework that uses general constraints on the application of Merge and the derivational timing of Spell-Out operations. Furthermore, I aim to demonstrate that the predictions made by the parser are actively mediated through immediate sensitivity to formal grammatical properties.

1.1 **Questions:** The research described in this proposal seeks to address the following questions: (i) during the real-time processing of Wh-question sentences, does the parser predictively build syntactic structures that contain PGs prior to receiving direct ("bottom-up") evidence that such gaps are present?; (ii) what is the time course during which the parser predictively builds syntactic representations?; (iii) is the parser sensitive to grammatical knowledge when building predicted structures?; (iv) at what point during a parse is grammatical knowledge applied?; (v) for sentences containing PGs, what are the structures that the parser builds – that is, what are the formal properties of PGs?; and (vi) what are the formal mechanisms that license sentences containing PGs?

1.2 **The Proposal:** The following proposal is divided into two parts. First, I will discuss an experimental program for investigating how the parser builds syntactic representations containing PGs. Secondly, I will describe a formal approach to characterizing the derivation of PG sentences.

Previous work (Phillips 2006) has demonstrated that subject PG sentences are well accepted by native speakers of English, and furthermore, that comprehenders will attempt to link an extracted Wh-phrase to a verb within a complex subject during real-time incremental processing. However, a question that remains is whether this subject-internal gap site is proactively predicted by the parser, or does the parser only posit a gap after it encounters the verb. For instance, earlier studies on cataphora resolution (Kazanina et al. 2007, Kush & Dillon 2021) have argued that the parser makes predictions about upcoming structural positions to avoid building representations what would violate binding constraints (Chomsky 1981). We can adopt a similar approach to test whether the parser makes predictions about PGs. For instance, if the parser has predictively built a structure containing a PG, then the parser could use this predicted gap to compute certain grammatical relationships; if we find evidence that such computations have taken place, then this would strongly suggest that the gap site was predicted. Let us consider a more concrete case by considering the example in (2), which is similar to the stimuli used in the proposed study 1.

(2) Which **father** did *his* plan to **{meet/eat}** __ inconvenience __?

Example (2) contains a PG contained within a complex subject, but how does the parser build this structure? From the findings reported in Phillips (2006), one possibility is that encountering direct evidence of the embedded verb (*e.g.*, *meet*) triggers the parser to posit a gap site. Whether a gap has been posited can be probed by comparing the reading times (RTs) at the verb for when the Wh-phrase is a good semantic fit for the argument position (*Which father…meet*) versus when it is a poor semantic fit (*Which father…eat*). When the Wh-phrase is a poor semantic fit with the verb, then attempting to link the Wh-phrase to a gap site following the verb is expected to result in comparatively slower RTs (Boland et al. 1995, Traxler & Pickering 1996). However, since the possessive pronoun matches in gender with the Wh-phrase (*father…his*), the parser is licensed to interpret the Wh-phrase as the pronoun antecedent. Having done so, linking the Wh-phrase to the gap site at the embedded verb would result in an ungrammatical strong crossover configuration, like shown in (3).

(3) a. *Which boy_i does he_i like $___i$?

b. *Which boy_i does his_i plan to meet $___i$ inconvenience __?

Therefore, examples like (2) provide us with cases in which a predicative parser is expected to behave differently from a non-predictive parser. That is, if the parser has predicted the embedded gap site prior to reaching the verb, then it can use this prediction to compute the strong crossover configuration and thereby stop the search for a gap within the subject. Evidence of this predictive behavior would be observed if there is no difference in RTs at the embedded verb, regardless of the semantic fit of the Wh-phrase, in just those cases where the intervening possessive and Wh-phrase are gender-matched.

The previous illustration outlines the general approach for experimentally investigating whether the parser predicts PGs. Using such designs, I will propose that the parser makes early predictions about the possible distributions of PGs in Wh-question sentences.

Next, I will roughly outline a formal approach for deriving PG sentences. In particular, I will be adopting a minimalist framework that accounts for PGs through sideward movement (Nunes 1995, Hornstein 2001) and Multiple Spell-Out (Uriagereka 1999b, Nunes & Uriagereka 2000). By applying this approach to complement clause PGs, prepositional dative PGs, and symbiotic gap constructions, I will additionally propose that the apparent anti-c-command property of PGs (Engdahl 1983) is an epiphenomenal consequence of sideward movement and not an independently necessary condition for the licensing of PGs. As a result, PG structures that violate anti-c-command, such as complement clause PGs and certain prepositional dative PGs, are proposed to arise through the non-application of sideward movement. I can then propose two classes on PG structures which can be characteristically distinguished based on whether the licensing gap c-commands the PG: (i) PGs incorporating sideward movement (subject and adjunct PGs) and (ii) PGs that do not incorporate sideward movement (complement clause and prepositional dative PGs). Such an approach however leaves open the question of how sentences like (4) are to be ruled out by the grammar.

(4) *Who __ filed the paper after reading __ ?

In example (4) the licensing gap c-commands the PG in the adjunct, and such examples have commonly been cited in the literature as an illustration of the anti-c-command condition on the distribution of PGs. Alternatively, I propose that examples like (4) can be accounted for in terms of locality constraints and legibility requirements imposed at the interfaces of the sort argued

for by Hornstein (2001). Therefore, broadly speaking, my proposal aims to reintroduce a *Barriers*-style analysis (Chomsky 1986) to constrain when Move can apply (c.f. Uriagereka 1999a for a separate attempt to integrate barriers into a minimalist framework). The end result is a characterization of Move that is generally free (barring locality constraints); movement is freely allowed into Theta-assigning positions and so-called *improper movement* (A' – A movement) is also possible. Once this description of Move is adopted, the PG structures attested in the literature fall out as a natural consequence. This proposed research program seeks to provide empirical support for this account of sentences containing PGs.

2. Study 1: Prediction of PGs during Real-Time Processing

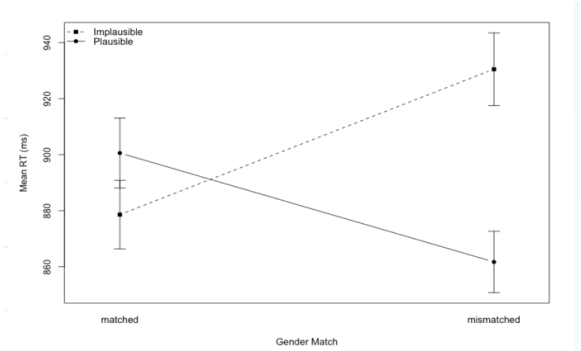
2.1 Experiment 1

This experiment is motivated by the reasoning outlined in section 1.2 and uses the design presented in (5) in which the gender match between the Wh-phrase and subject-embedded possessive pronoun is crossed with the goodness of semantic fit of linking the Wh-phrase to the embedded verb.

(5) Which new **{father/mother}**, as mentioned by several onlookers, did *his* sudden plan to quickly **{meet/eat}** in the restaurant with the busy doctor greatly inconvenience __?

As described previously, if the parser predicts a gap within the subject, then encountering a pronoun that co-refers with the Wh-phrase would result in a strong crossover configuration and potentially lead the parser to reanalyze its prediction to avoid building an ungrammatical structure. If so, we would only expect an effect of semantic fit at the embedded verb in just those cases where the Wh-phrase and possessive pronoun are gender-mismatched. In a pilot study using 150 participants, sentences like (5) were presented in a lexical maze (L-maze) task (Witzel et al. 2012). The results of the study confirmed the hypothesis that the parser predicted a gap in the subject prior to reaching the embedded verb. Crucially, we detected a gender-mismatch effect at the pronoun such that when the pronoun mismatched in gender with the Wh-phrase, RTs were slower. This finding indicates that comprehenders did attempt to interpret the Wh-phrase as the pronoun antecedent but were forced to reanalyze when a gender mismatch was present. More importantly, we found a significant interaction at the embedded verb showing that RTs were moderated by the gender match of the intervening pronoun, as shown in (6). This result clearly shows that the parser did not posit a gap at the verb when doing so would lead to an ungrammatical strong cross-over configuration. This suggests that the parser had predicted the gap, computed the presence of strong cross-over, and subsequently reanalyzed its prediction to avoid this predicted ungrammaticality.

(6) Average RT (ms) at the subject-embedded verb



Two additional conclusions about the processing of PGs are suggested by the finding. First, this result is broadly incompatible with proposals that seek to account for the location of

gaps inside of PG domains in terms of processing limitations (Hofmeister & Sag 2010). For instance, Chaves (2013) argues that the findings reported by Phillips (2006), in which gaps were posited inside of infinitival subjects but not finite ones, may be a consequence of increased processing difficulty associated with locating gaps inside of finite subject domains. That is, the parser is more-or-less free to posit gaps inside of subjects, however doing so incurs a processing cost. If this processing cost is high enough, then working memory becomes "overloaded" and the parser's search for a possible gap site is suspended. Therefore, a general correlation between increased processing difficulty and the cessation of gap search operations is predicted. However, the findings of our pilot show the opposite pattern: we found evidence that the parser was actively positing gaps in only the comparatively more difficult conditions that involved a gender-mismatch. This finding offers support for Phillips' (2006) conclusion and suggests that processing inside of PG domains is crucially determined by grammatical considerations rather than general processing limitations.

Secondly, this pilot study provides some evidence about the timing of the parser's prediction mechanism. Kush et al. (2017) showed that the parser avoids linking a pronoun with an extracted Wh-phrase when doing so would result in a strong crossover configuration with a predicted gap in the matrix clause. However, when the pronoun was embedded as a possessive, the possessive pronoun was interpreted as coreferential with the Wh-phrase. From this finding the authors concluded that a matrix gap is predicted as soon the Wh-phrase is encountered, and that the parser is sensitive to strong crossover configurations but not to weak crossover configurations. In our pilot study, the parser is essentially "tricked" into establishing a coreference relationship between the possessive pronoun and the Wh-phrase that would result in a strong cross-over configuration. Based on the results of Kush et al. (2017) and our pilot study, we can therefore conclude that the parser does not posit a subject PG immediately upon encountering the Wh-phrase, nor does it posit one when it integrates the possessive pronoun into the syntactic representation. Instead, our findings suggest that the parser posits a subject PG at some point after encountering the possessive pronoun but prior to reaching the embedded verb. A reasonable conclusion is that the parser posits a subject PG as soon as it detects that a complex subject is present, possibly at the word *to*.

2.2 Experiment 2 Previous work by Wagers & Phillips (2009) shows that after locating a gap in the matrix clause, the parser continues active gap search into conjuncts but not into adjuncts. The authors concluded that active gap search is determined by grammatical requirements rather than by maximal interpretation. That is, the parser searches for a second gap inside of conjuncts because failing to locate one would violate the coordinate structure constraint, whereas a PG inside of adjunct clause, while possible, is not necessary for the resolution of the Wh-dependency. Evidence of active gap search inside of subject domains (Phillips 2006) complicates this picture, since locating a gap inside the subject does not resolve the open dependency. One possibility is that upon encountering a Wh-phrase, the parser predicts a gap inside of the matrix clause and, so long as this initially predicted gap remains unfilled in the syntactic representation, will actively search for a gap in any domain where one can grammatically occur. In this sense, the initial gap prediction acts as a 'trigger' for an active search that remains switched on until the gap is filled. To test this hypothesis, I propose an experiment that examines the processing of sentences containing fronted adverbial PGs, as in (7).

(7) a. These are the employees who, after **{meeting/eating}** with them, the manager criticized __.

b. These are the employees who the manager criticized __ after **{meeting/eating}** with them.

The example item illustrated in (7) uses a 2*2 design that manipulates the position of the adjunct phrase (fronted/non-fronted) and the semantic fit of linking the relativized NP with the verb inside the adjunct. If the parser actively searches for a gap inside of the adjunct, then we would expect longer RTs at the adjunct verb when the relativized NP is a comparatively poor semantic fit with the verb (e.g., *employees…eating > employees…meeting*). Based on the results reported by Wagers & Phillips (2009), we predict no effect of semantic fit for (7b), however (7a) presents the crucial comparison. One possibility is that the lack of active gap search that Wagers & Phillips reported reflects a general property of adjunct processing – the parser simply does not look for gaps inside of adjunct domains. Alternatively, if active gap search is a mechanism that is triggered by the predication of a matrix gap and is only switched off when this gap is conclusively located, then we would expect to see an effect of semantic fit in (7a) but not in (7b). I therefore propose to conduct an L-maze task that tests items like (7) to further explore how the parser builds structures inside of PG domains.

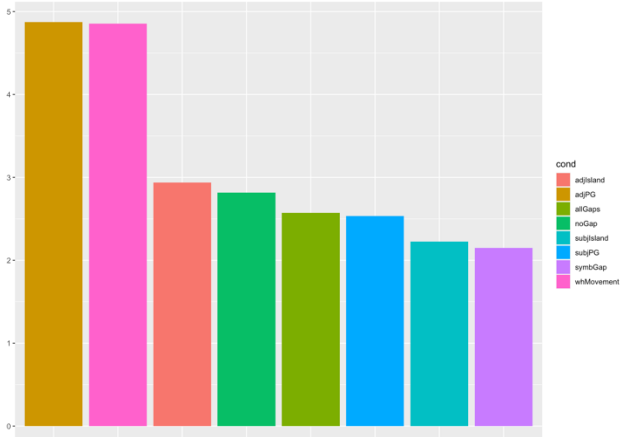
3. Study 2: Formal Properties of PG Structures

3.1 Experiment 3

Levine & Sag (2003) have argued that null-operator accounts of parasitic gaps (Chomsky 1986, Nissenbaum 2000) fail to account of the acceptability of symbiotic gap structures (1e). Such structures are similarly not predicted by the sideward movement analysis pursued in this proposal. We therefore conducted a pilot study to determine the acceptability of symbiotic gaps. In an acceptability judgement task, we used a 2*2*2 design that systematically manipulated the presence of a gap in the subject, adjunct, and matrix. An example item set is given in (8). The results of this task using 80 participants are shown in (9).

(8) What kinds of books do authors of {science textbooks/__} hate editing {the introduction/__} after writing {the first draft/__}?

(9) Acceptability of Multi-gap sentences



There are two main findings from the result of this pilot study. First, contrary to claims made by Levine & Sag (2003), English speakers appear to find sentences containing symbiotic gaps to be generally unacceptable – of all the sentences tested, the symbiotic gap condition was found to be the least acceptable. More importantly, we found that the presence of symbiotic gaps did not improve the acceptability of sentences containing subject or adjunct island violations.

Symbiotic gap sentences were rated as similarly unacceptable as subject island violating sentences and significantly worse than adjunct island violating sentences. Based on this finding, we can therefore set aside symbiotic gaps as a phenomenon in need of explanation. A further finding however, is that the "all gaps" condition in which a gap is present in the subject, adjunct, and matrix conditions was rated similarly to subject PG sentences. This finding is important, as the acceptability of this condition is predicted by the sidewards movement with Multiple Spell-out account of PGs. The derivation is as follows: first the adjunct clause is constructed through the repeated application of Merge. Then, prior to spelling out the complete adjunct, the Wh-phrase positioned as the argument of *writing* is spelled-out, copied, and sideward moved into the argument position of the matrix verb (*editing*). This movement is triggered as a last resort to satisfy the selectional requirements of the matrix verb. At this point, the complete adjunct is spelled-out and merged to the matrix VP. Subsequently, in a separate tree the complex subject is constructed, however once again there is no object present that can satisfy the selectional requirements of *of.* Since the adjunct clause has been spelled-out, the copy of the Wh-phrase that was originally merged there is no longer available to be sideward moved into the subject. However, the Wh-copy in the matrix clause remains available, and as a last resort it is copied and sideward moved into the subject. The full subject is then spelled-out and merged into the sentence spine. The derivation then proceeds as normal until the C head at the left periphery of the sentence is merged in. The C head presumably carries with it a Q feature that must be locally checked by a Wh-phrase in the Spec, C position. As before, since the adjunct clause and complex subject have been spelled-out, the Wh-copies located these domains cannot be targeted for movement. Critically, the Wh-copy positioned as the argument of *editing* continues to be available, and it is copied a second time and moved into Spec, C. Finally, the full sentence is spelled-out.

If we compare this derivation with the one that a sideward movement with Multiple Spell-out account predicts for symbiotic gaps, we see that the derivation crashes when constructing the complex subject. Specifically, since there is no gap in the matrix, the full adjunct is spelled-out and merged into the spine without the Wh-phrase being copied and moved. As a result, when the complex subject is being built, there is no object visible to syntax that can be used to satisfy the selectional requirements of *of*. This result follows if we assume two additional constraints on sideward movement and Multiple Spell-out: (i) objects may not be copied and held in the workspace indefinitely – copied objects must be immediately merged; and (ii) there are limitations on the number of trees that can be concurrently "active" in the workspace. This latter constraint has been previously argued for in the minimalist literature (c.f., Nunes & Uriagereka 2000). Therefore, not only does a sideward movement with Multiple Spell-out analysis make the correct prediction about the impossibility of symbiotic gaps, it also predicts the availability of triple gap structures that our pilot study indicates are similarly acceptable as subject PG sentences. It is not clear how a null-operator movement analysis would account for this triple gap sentence.

3.2 Experiment 4 Recent work by Arregi & Murphy (2022) has argued for a null-operator movement analysis to account for complement clause PGs (1d). Their analysis crucially requires anti-c-command as a necessary condition for the licensing of PGs, and they therefore suggest that the complement clause must extrapose to a non-c-commanded position. In support of this claim, they present the following contrast in (10).

(10) Who did you persuade __ [??(that) we should visit __] ?

Arregi & Murphy argue that the acceptability of (10) degrades when the complementizer is omitted. They account for this degradation by appealing to a suggestion made by Stowell (1981) that complementizers must be properly governed and that this is only possible if the complement clause is positioned as a verb argument. This contrast is not predicted by the analysis proposed in the current paper and therefore an alternative account of complement clauses is required. Importantly, it should be noted that the contrast in (10) seems to be fairly subtle and it has been suggested that depending on the matrix verb the contrast may disappear entirely (Contreras 1984). I therefore propose an acceptability judgement task using the design given in (11) in which the presence of the complementizer is crossed with the location of an adverbial.

(11) a. Who did you persuade __ yesterday (that) we should visit __?

b. Who did you persuade __ (that) we should visit __ yesterday?

I predict that omitting the complementizer only leads to decreased acceptability when extraposition is forced by the presence of an intervening adverbial, as in (11a). Conversely, if the absence of the complementizer makes no significant difference to the acceptability of (11b), then this casts doubt on the claim that complement clause PGs require extraposition, and further suggests that anti-c-command is not a requirement for the licensing of PGs. Under this analysis, the Wh-phrase inside the complement clause moves successive cyclically to the embedded Spec, C position followed by subsequent movement into an argument position of the matrix VP. Like with sideward movement, this application of Move would violate the Theta-criterion. It would additionally constitute an instance of so-called improper movement. As such, the current analysis requires that these two conditions be reformulated or removed from the grammar, a position that has been previously argued for in the literature (Hornstein 2001).

3.3 Experiment 5 Arregi & Murphy additionally report the following contrast:

(12) a. Which girl did you give a picture of __ to __ yesterday?

b. *Which book did you give __ to a fan of __ yesterday?

The sentences in (12) are examples of prepositional dative PGs, and Arregi & Murphy argue that the unacceptability of (12b) derives from the violation of anti-c-command. Whereas in (12a), the higher gap is embedded inside of a larger NP such that anti-c-command is satisfied. The current analysis aims to reduce anti-c-command to an epiphenomenal effect of sideward movement. Therefore, if sideward movement is not involved in the derivation of (12b), then the claimed unacceptability is unexpected. To test the effect of c-command on PGs, I propose the following acceptability judgement task that compares complement clause PG sentences in which the higher gap is positioned in the subject position (13a) or in the dative position (13b). As a baseline, each condition will be compared against sentences in which all gap positions are filled.

(13) a. Who did John introduce __ to an admirer of __ yesterday?

b. Who __ introduced John to an admirer of __ yesterday?

Under the account pursued by Arregi & Murphy (2022), both (13a) and (13b) should be bad for the same reason – they violate anti-c-command. Therefore, if we observe a contrast between the conditions, then this suggests that anti-c-command is not driving the unacceptability. Instead, I propose that the degraded acceptability of (13a) relative to (13b) arises out of the violation of locality conditions – the subject position is “too far” for the Wh-phrase to move to

it. The informants I have consulted have generally been in agreement; both sentences are degraded, however they may be degraded for different reasons. Most tellingly, my informants claim that it is impossible for them to link the Wh-phrase to the lower gap position in (13b), whereas (13a) seems to allow for parasitism but the resulting question is highly difficult to interpret. If this intuition is on the right track, then the contrast reported in (12) might possibly be due to extra-syntactic reasons. In any case, the proposed design in (13) can help clarify the role, if any, that anti-c-command has in the licensing of PGs.

4 Summary Two studies have been described in this research proposal. The first study uses the L-maze task to investigate how sentences containing PGs are processed in real-time. In the second study, acceptability judgement tasks are proposed in support of formal account of PGs that relies on the availability of sideward movement and Multiple Spell-out. If we reconsider the research questions raised earlier in this proposal, then pending the results of the proposed experiments, I hope to provide the following answers:

(14)	Q: Does the parser predict PGs?	A: Yes, so long as the matrix gap is unfilled.
	Q: When are PGs predicted?	A: After the parser recognizes the parasitic domain.
	Q: Sensitivity to grammar?	A: Predictions are discarded if ungrammatical.
	Q: Time course of sensitivity?	A: Grammatical knowledge is applied immediately.
	Q: Structure of PGs?	A: First Wh-copy is moved to each gap position.
	Q: Formal mechanisms?	A: Sideward movement with Multiple Spell-out.

References Cited

Arregi, Karlos & Andrew Murphy. 2022. Argument-internal parasitic gaps. Ms. University of Chicago.

Boland, Julie E., Michael Tanenhaus, Susan Garnsey, & Greg N. Carlson. 1995. Verb-argument structure in parsing and interpretation: Evidence from WH-questions. *Journal of Memory and Language, 34*, 774–806.

Contreras, Heles. 1981. A note on parasitic gaps. *Linguistic Inquiry 15*(4), 698-701.

Chaves, Rui. 2013. An expectation-based account of subject islands and parasitism. *Journal of Linguistics 49*(2), 285-327.

Chomsky, Noam. 1981. *Lectures on government and binding.* Berlin: Mouton de Gruyter.

Chomsky, Noam. 1986. *Barriers*. MIT Press, Cambridge, MA.

Chomsky, Noam. 1995. *The minimalist program.* Cambridge, Mass.: MIT Press.

Engdahl, Elisabet. 1983. Parasitic gaps. Linguistics and Philosophy 5, 5–34.

Hofmeister, Phillip & Ivan Sag. 2010. Cognitive constraints and island effects. *Language 86*(2), 366-415.

Hornstein, Norbert. 2001. *Move! A minimalist theory of construal*. Mass.: Blackwell Publishers.

Kazanina, Nina, Ellen Lau, Moti Lieberman, Masaya Yoshida and Colin Phillips. 2007. The effect of syntactic constraints on the processing of backwards anaphora. *Journal of Memory and Language 56,* 384-409.

Kush, Dave & Brian Dillon. 2021. Principle B constrains the processing of cataphora: Evidence for syntactic and discourse predictions. *Journal of Memory and Language, 120*.

Kush, Dave, Jeffrey Lidz, & Colin Phillips. 2017. Looking forwards and backwards: The real-time processing of strong and weak crossover. *Glossa 2*(1), 1-29.

Levine, Robert D. & Ivan A. Sag. 2003. Some empirical issues in the grammar of extraction. In Stefan Müller (ed.), *Proceedings of the 10th International Conference on Head-Driven Phrase Structure Grammar.* Michigan State University, 236–256. DOI: 10.21248/hpsg.2003.14.

Nissenbaum, Jon. 2000. *Investigation of covert phrase movement.* Ph.D dissertation, MIT.

Nunes, Jairo. 1995. *The copy theory of movement and linearization of chains in the minimalist program.* Ph.D. dissertation, University of Maryland, College Park.

Nunes, Jairo & Juan Uriagereka. 2000. Cyclicity and extraction domains. *Syntax, 3*(1), 20-43.

Phillips, Colin. 2006. The real-time status of island phenomena. *Language 82*, 795-823.

Stowell, Tim. 1981. *Origins of phrase structure.* Ph.D dissertation, MIT.

Traxler, Matthew J. & Martin J. Pickering. 1996. Plausibility and the processing of unbounded dependencies: An eye-tracking study. *Journal of Memory and Language, 35*, 454–475.

Uriagereka, Juan. 1999a. Minimal restrictions on Basque movements. *National Language and Linguistic Theory 17*, 403-444.

Uriagereka, Juan. 1999b. Multiple spell-out. in Samuel D. Epstein and N. Hornstein (eds.) *Working Minimalism*. MIT Press: Cambridge, MA, 251-82.

Wagers, Matthew & Colin Phillips. 2009. Multiple dependencies and the role of the grammar in real-time comprehension. *Journal of Linguistics 45*(2), 395-433.

Witzel, Naoko, Jeffrey Witzel, & Kenneth Forster. 2012. Comparisons of online reading paradigms: Eye tracking, moving-window, and maze. *Journal of Psycholinguistic Research 41,* 105–128.